

Architectural Particular Specification
METAL DOORS
SECTION 7.1

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SECTION 7.1

METAL DOORS

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The Work of this Section includes, but is not limited to, the furnishing and installing of the following:
1. All fire rated and non-rated Steel doors & frames.
 2. Beads, trim, seals and all other associated accessories
 3. Fire test and assessment of fire-rated door assemblies
 4. Proprietary acoustic steel door according to Specification of Acoustic Performance of Acoustical Doors under this Contract.

1.02 REFERENCES

- A. British Standards
1. BS 476: Part 20: 1987 (1990) - Method of Determination of the Fire Resistance of Elements of Construction.
 2. BS 476 :Part 22 : 1987 - Method of Determination of the Fire Resistance of Non-loadbearing Elements of Construction.
 3. BS 476: Part 23: 1987 - Method of Determination of the Contribution of Components to the Fire Resistance of a Structure.
 4. BS 476: Section 31.1: 1983 - Method of Measuring Smoke Penetration Through Doorsets and Shutter Assemblies Under Ambient Temperature Conditions.
 5. BS 970 : Part 1 : 1991 : General Inspection and Testing Procedures and Specific Requirements for Carbon, Carbon Manganese, Alloy and Stainless Steels.
 6. BS 729 : 1971(1986) : Specification for Hot Dip Galvanized Coatings on Iron and Steel Articles.
 7. BS 5872 : 1980 : Specification for Locks and Latches for Doors in Buildings.
 8. BS 5466 : Part 7 : 1982(1990) : Guidance on Stationary Outdoor Exposure Corrosion Tests.
 9. BS 6510: 1984 - Specifications for Steel Windows, Sills, Window Boards and Doors.
 10. BS 1245: 1975 - Metal Door Frames (Steel).
 11. BS 8220: Part 2: 1995 - Guide to Security of Building Against Crime: Offices and shops.
 12. BS DD 171: 1987 - Guide to Specifying Performance Requirements For Hinged or Pivoted Doors.
 13. PD 6512: Part 1: 1985 - Use of Elements of Structural Fire Protection with Particular Reference to the Recommendation Given In BS 5588: Guide to Fire Doors
- B. American Standards
1. ANSI A224.1-1990: Standard Test Procedure and Acceptance Criteria for Prime Painted Steel Surfaces for Steel Doors and Frames.
 2. ANSI A250.3-1993: Performance Test Procedure and Acceptance Criteria for Prime Painted Steel Surfaces for Steel Doors and Frames anchors.

3. ANSI A250.4-1994: Test Procedure and Acceptance Criteria for Physical Endurance for Steel Doors and Hardware Reinforcing.
4. ANSI/NFPA 252-1990: Fire Tests of Door Assemblies.
5. ANSI/NFPA 255-1990: Surface Burning Characteristics of Building Materials.
6. UL 1785 - 95 : Air Leakage Tests for Door Assemblies
7. ASTM E152 - 81a : Method of Fire Tests of Door Assemblies (Hose Stream Test)
8. ASTM E283-84: Test methods of Air Leakage Through Exterior Windows, Curtain Wall and Doors.
9. American Wood Work Institute: AWI Quality Standard for Architectural Flush Doors.
10. ASTM B117 -94: Salt Spray (Fog) Testing (Paint Test).
11. ASTM D1735-92: Method of Water Fog Testing of Organic Coatings.
12. ASTM E283-73 : Door Air Infiltration Test
13. NFPA 105-89: Installation of smoke control Door Assemblies.
14. NFPA 80 : Fire Doors and Windows (current edition).
15. All fire rated doors shall meet the stability, integrity and insulation requirements of BS 476 : Parts 8, 20 and 22 ANSI/NFPA 252 and 253. All fire shall be fire tested and approved with the specified door seals, ironmongery and metal door together as an integral assembly. Any deviations in the door and frame construction, material, ironmongery type, and method of fixing / installation shall be assessed by an recognized Testing Agency.
16. All door fire seal, smoke seal and weather strippings shall comply with the following requirements:
 - a. Fire Test Certificate and Approval
 - i) All assemblies of fire rated doors and frames are to be identified by the SISIR label.
 - ii) All door frames materials shall have been tested, approved and listed by U.S. Underwriter's Laboratory (UL), material capable of passing ASTM E152-81a, ASTM E283-73, NFPA 105-89, NFPA 80, and UL 1784.
 - iii) Alternatively, all firestop and barrier seal system materials shall be certified by the British Board of Agreement (BBA), tested to BS 476 : Part 20 and Part 22, and all referenced Standards cited herein.
 - iv) All door fire seals and smoke seals shall be "CERTIFIRE" and BBA approved, or equivalent UL-labeled seals.
 - v) When tested to BS 476 : Section 31.1, air infiltration/exfiltration shall not exceed 7m³/h or 0.25m³/h per metre of any door perimeter air sealing, whichever is greater, at pressure of 50Pa above atmospheric pressure.
 - vi) All door smoke seals shall be tested to BS 476 : Section 31.3 and NFPA 105-89 to restrict the passage of hot smoke to a maximum 6% obscuration per foot (19% obscuration per metre).

1.03 QUALITY ASSURANCE

- A. Qualification of Steel Door Manufacturer: Company specializing in the manufacturing of high-quality, commercial-grade, fire rated steel doors with minimum of ten (10) years documented experience (most recent five (5) years prior to Tender date).

1.04 PERFORMANCE REQUIREMENTS

- A. The design of doors and louvres shall comply with the requirements stipulated in the latest revisions of the :
1. Code of Practice on Wind Effects Hong Kong.
 2. Building (Construction) Regulations.
 3. Building (Planning) Regulations.
 4. Buildings Ordinance (Chapter 123).
 5. British Standards referenced herein.
- B. Fire & Smoke Protection Requirement for Architectural Flush Doors : Premium Grade Construction:
1. All fire-rated steel doors shall be fire-tested and certified to rating up to 240/240/240 FRR. All fire-rated steel doors shall have non-removable fire labels mounted on the top of the door.
 2. All fire doors shall have the ability to resist burn-through and remain free from critical levels of distortion under fire exposure.
 3. Any voids due to over mortising as a result of fitting ironmongery shall be filled with suitable weather proof intumescent fire stop material. All gaps or voids around door edge lippings, around apertures and within core material shall be filled with suitable weather proof intumescent fire stop paste.
 4. All fire doors shall meet the stability, integrity and insulation requirements of BS 476: Parts 20 and 22 ANSI/NFPA 252 and 253. All fire doors shall be fire tested and approved with the specified door seals, ironmongery and metal door together as an integral assembly. Any deviations in the door and frame construction, material, ironmongery type, and method of fixing/installation shall be assessed by an recognized Testing Agency.
 5. All door fire seal, smoke seal and weather strippings shall comply with the following requirements:
 - a. Fire Test Certificate and Approval
 - vii) All assemblies of fire rated doors and frames are to be identified by the SISIR label.
 - viii) All door frames materials shall have been tested, approved and listed by U.S. Underwriter's Laboratory (UL), material capable of passing ASTM E152-81a, ASTM E283-73, NFPA 105-89, NFPA 80, and UL 1784.
 - ix) Alternatively, all firestop and barrier seal system materials shall be certified by the British Board of Agreement (BBA), tested to BS 476 : Part 20 and Part 22, and all referenced Standards cited herein.
 - x) All door fire seals and smoke seals shall be "CERTIFIRE" and BBA approved, or equivalent UL-labeled seals.
 - xi) When tested to BS 476 : Section 31.1, air infiltration/exfiltration shall not exceed 7m³/h or 0.25m³/h per metre of any door perimeter air sealing, whichever is greater, at pressure of 50Pa above atmospheric pressure.
 - xii) All door smoke seals shall be tested to BS 476 : Section 31.3 and NFPA 105-89 to restrict the passage of hot smoke to a maximum 6% obscuration per foot (19% obscuration per metre).

6. When tested to BS 476 : Section 31.1, air infiltration/exfiltration shall not exceed 7m³/h or 0.25m³/h per metre of any door perimeter air sealing, whichever is greater, at pressure of 50Pa above atmospheric pressure.
7. All door smoke seals shall be tested to BS 476: Section 31.3 and NFPA 105-89 to restrict the passage of hot smoke to a maximum 6% obscuration per foot (19% obscuration per metre).
8. Protective Coating Requirements
9. All door frames and frame components shall be factory cleaned, phosphatized and finished as standard with one coat of baked on rust inhibiting primer paint in accordance with ANSI A224.1.

C. Acoustical Performance

1. The Contractor shall supply and install all acoustic doors to achieve the acoustic ratings in the specified areas or as indicated on the Contract Drawings or Schedules.
2. Read in conjunction with Specification of Acoustic Performance of Acoustical Doors and Specification of Acoustic Performance of Entry Doors.
3. The Contractor shall not install acoustic doors without prior approval from the Architect.
4. The doors shall be assembled in factory completed with specified acoustic seals and hung in frame for easy site installation. All doors shall be of steel construction as specified.
5. For fired rated door with STC 35 rating, door leaf and frame shall be constructed of stainless steel shall be between 45 to 55mm, or as otherwise specified.

For proprietary acoustic rated metal door with STC 50, door leaf and frame shall be constructed of stainless steel shall be between 45 to 60mm filled with sound absorbing material and damping elements, or as otherwise specified.
6. Door hinges shall be high quality stainless steel ball-bearing lift off hinges fixed to adjustable mounting in door leaf and frames in factory. The hinge mechanism shall be tested in factory prior to delivery to ensure a smooth door operation. Manufacturer to furnish laboratory test data certifying that hinges of identical design have been cycled a minimum of 125,000 times while supporting a door leaf weighing a minimum of 350 lbs. (158.8 kg).
7. The doors shall be fitted with all necessary ironmongery including locks and handles. There shall be no through key holes, unless otherwise specifically approved by the Architect in writing. The Contractor shall ensure that the said locks will not affect the performance of the doors.
8. Where acoustic seals are specified, ironmongery shall be selected to ensure that all of the perimeter is evenly compressed, with adequate pressure to provide an effective acoustic seal, without over-compression that may result in difficulty of door closure or seal distortion. All acoustical doors shall be installed with perimeter seals meeting test requirements of ASTM E90-90 & ASTM E1408-91.
9. For doors requiring STC 50 rating the side and head of door and frame shall receive two (2) sets of self-aligning magnetic compression seals while for doors requiring STC 35 rating shall receive one (1) set of self-aligning magnetic compression seals. Door to be held in closed position by magnetic force of perimeter seals. Acoustic labyrinth shall be created when door is in closed position. Bottom of door leaf shall contain continuous gravity-activated seal which shall compress against floor as door is closed. Raised sills, threshold, drop seals, and mullions (pairs only) will not be permitted.

10. Doors for Conference Rooms and other specified areas, shall have a minimum STC rating of 50, installed and be installed with fully rebated acoustical seals when tested in accordance with ASTM E90-85 of not less than the following:

Octave Band Centre Frequency, Hz	63	125	250	500	1K	2K	4K	8K	STC
Sound Transmission Loss, dB	22	27	41	46	50	52	54	56	50

D. Physical Endurance Performance Requirements:

1. All steel doors shall be Grade II Heavy-Duty Level capable of meeting the door tests of BS DD 171: 1987 for impact, torsion and closing against an obstruction tests.
2. All door frames shall be Grade II Heavy-Duty Level capable of meeting the door tests of ANSI A250.4-1994 with 1 million swing operating cycles and twist tested 23 times at regular intervals during the cycle test.
3. After completion of ANSI A250.3 test, no metal fatigue, cracking or deformation at hardware provisions cut outs or along form contours shall occur. No weld breakage and misalignment shall occur.
4. All door frames and frame anchors shall be tested and meet the requirement of ANSI A250.5-1994.
5. After completion of BS DD 171 test, no structural failure warping exceeding 3 mm, or deformation at hardware provisions cut outs shall occur. Submit test results to the Architect for review.

E. Resistance To Wind Load

1. Design wind load for doors and louvres shall be calculated in accordance with the latest revision of the Code of Practice on Wind Effects Hong Kong, subject to a minimum design wind load of 3kPa for positive pressure or 4.2 kPa for negative pressure.
2. Doors and louvres when fixed in position are to withstand the design wind load with permissible maximum deflection of less than 1/240th of the span length or 12mm in individual members, whichever is less.
3. In the plane of the wall, deflection of any glazing member shall not reduce the glass bite below 75% of the design dimension, or 3mm whichever is greater. Restrict deflection further if required for proper assembly and fit of all components. The glass panel edge clearances shall not be less than 25% of the design dimension or 3mm, whichever is greater.
4. At connection points of framing members to anchors, including cantilever braces, kickers etc., anchor deflection in any direction shall not exceed 1mm.
5. Where required to limit deflection and maintain structural integrity under wind load, use galvanised mild steel cores, anchors etc. as required to provide the necessary stiffness.

F. Thermal Movement

1. Doors and louvres shall be designed and installed to accommodate thermal expansion and contraction in the frame and glazing under local climatic and site conditions. The manufacturer shall supply full details of the thermal design.
2. The thermal design shall be based upon a minimum material temperature increase and decrease of 55°C relative to nominal condition. In no case shall the glass bite be less than 10mm or edge clearance less than 6mm.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

- A. The Contractor and Door Manufacturer shall be aware of the Indoor psychometric perimeters used by the Consulting Mechanical Engineer in determining vapour barrier, insulation, indoor temperature and relative humidity requirements, heating and cooling loads.
- B. The Contractor and Timber Door Manufacturer shall warrant that all products, materials and items are suitable, and can be maintained in the various climatic and indoor psychometric conditions encountered without warpage, face de-lamination, material deterioration and other defect.

1.07 SUBMITTALS

- A. Propose manufacturer specializing in the production of premium grade steel fire door sets with ten (10) years minimum experience on projects of a similar size and complexity.
- B. Shop Drawings:
 - 1. Indicate profiles, sizes, connection attachments, reinforcing, anchorage, core and edge construction.
 - 2. Include Sections and elevations in 1:20 scale and details in 1:2 scale.
 - 3. Indicate structural opening dimensions and door opening dimensions, locations, length & type of all intumescent and smoke seals and cut outs. Show details.
 - 4. Indicate surface weight of door leaf including all accessories and ironmongery installed to achieve the specified door acoustic rating.
 - 5. Indicate welded connections using standard welding symbols. Indicate net weld length.
- C. Samples:
 - 1. Minimum 600 mm x 600 mm size samples of part of each door panel showing core construction, edge treatment, face materials, frames and vision panel.
 - 2. Provide a 600 x 600 sample of the door and door frame assembly with specified hinges, lock, handles and all required smoke and intumescent seals fitted and installed.
- D. Catalogue of proposed acoustic door seals
- E. Submit fire and acoustic test certificate and assessments for each type of fire door assembly.
- F. Submit at least six field tests reporting a rating of not less than the required Noise Isolation Class. Tests shall be carried out on doors, which have been installed less than one year.

- G. The Contractor shall submit acoustic test certificates to demonstrate compliance with the specified acoustic rating for the proposed acoustic doors. Where valid acoustic test certificates are not available, the Contractor shall, at his own cost, conduct necessary acoustic tests to demonstrate compliance. All acoustic tests shall be performed by an independent recognized acoustic laboratory to be approved by the Architect.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY, STORAGE AND HANDLING

- A. Deliver and store on project Site all steel doors and frames only when the Project Building is completely enclosed, moisture-laden materials such as concrete, plaster, etc. have largely dried, and heating or cooling system are functioning to maintain uniform indoor psychometric conditions.
- B. Protect all steel doors from damage, staining, traffic and Work of other trades. Ensure that all door edges, frames and corners are protected from chipping or other damage.
- C. All door frames shall be braced with adjustable base ties to hold frames rigid during transit and erection. Base ties shall be fitted to suit thickness of floor finish. Remove base ties after installation.
- D. Door leaves shall be stored flat on leveled 100 x 50 wood sleepers on length equal to full width of doors. Support sleepers shall be covered with clean cardboard to prevent marking of the bottom door.
- E. Ensure that all fire doors are affixed with Fire test/approval certificate metal labels, located in a position acceptable to the Architect.
- F. Never stand panels on end against a wall for any length of time. If it is not convenient to lay them flat, they shall stand on their sides with care taken not to chip the sides and edges. Protect all edges and corners from damage.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

- A. All assemblies of fire rated doors and frames shall have test certificate(s) to demonstrate the whole installation is acceptable to the Buildings Department (BD) and Fire Service Department (FSD), with regard to the specified Fire Resistance Rating.
- B. All door frames materials shall have been tested, approved and listed by U.S. Underwriter's Laboratory (UL), material capable of passing ASTM E152-81a, ASTM E283-73, NFPA 105-89, NFPA 80, and UL 1784.

- C. Alternatively, all firestop and barrier seal system materials shall be certified by the British Board of Agrément (BBA), tested to BS 476 : Part 20 and Part 22, and all referenced Standards cited herein.
- D. All door fire seals and smoke seals shall be “CERTIFIRE” and BBA approved, or equivalent UL-labeled seals.

1.12 WARRANTY

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.
- B. The Contractor and Steel Door Manufacturers shall jointly provide warranty against material defects, fire protection and workmanship for Work of this Section.
- C. Material defects shall include, but not be limited to warpage, sagging, deterioration, delamination or defects, during the warranty period.
- D. Nothing shall be done by the Contractor which will void any Manufacturer's Warranty, or invalidate the Project Building's Loss-Prevention Insurance coverage.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 MATERIAL

A. Stainless Steel Faced Door:

1. Shall be of a flush type, with seamless edges.
2. All steel louvers and doors shall be manufactured from :-
 - a. Hot rolled steel sections,
 - b. Cold straightened,
 - c. Free from rolling defects, and
 - d. Generally in accordance with B.S. 6510 : 1984.
3. Door leaf skins shall be fabricated from 1.5mm Stainless Steel Type 316, S11 austenitic, satin finish with vertical galvanised steel V shaped stiffeners at minimum 200mm center-to-centres and internal galvanised steel reinforcement at all ironmongery positions. The minimum thickness of the reinforcement is to be 1.5mm.
4. All doors shall be fully filled with insulation of incombustible mineral wool with a minimum density of 150kg/cm³. The core materials shall meet the general requirements of section 07210 together with the specified door performance requirements.
5. It is to be noted that all ironmongery, smoke, intumescent and weather seals are to be fully rebated into the door and frame construction.
6. The minimum door leaf weight is to be 26 Kg/m².
7. Double leaf doors shall have a folded stainless steel (min. thickness 1.5mm) astragal, with overlap at the pull side, at the push side the leaf should be seamless.

B. Exterior and Interior Flush Door Frames :

1. Frames to be provided together with doors as complete door sets.
2. Square and flat,
3. Constructed with mitred, welded corners,
4. With glazing bars machine tenoned or welded to frames,
5. Drilled to receive glazing clips, fixing lugs and fixings as required.
6. Door frame jamb depth, profile and rebate vary. Refer to the Drawings for each location and application. 30mm minimum door frame stop for fire-rated doors.
7. Provide mortar guards welded to door frame where cut-outs are provided for lock strike plates or bolts. Door frames for single doors shall be fitted with an adjustable lock strike plate, zinc coated sheet complying with BS491 or BS 1706, Classification ZN3.
4. Reinforcement plates of 2.5mm thick minimum shall be installed on all door frames for installation of door closers, hinges, locks, and panic hardwares. Install L-shape frame corner reinforcements at head-jamb transitions.
5. Interior Door Frames: Formed stainless steel grade 316, S11 austenitic of 1.5mm thickness (16 gauge) per BS 1449: Part 2 : 1983 and BS 4360. Provide with all corners mitred and welded. All welds to be ground smooth. Hot-dip galvanized steel fixing bracket complying with BS 2989 class 2A and treated with 1 coat of calcium plumbate priming paint complying with BS 3698.

Exterior Door Frames: Formed stainless steel grade 316, S11 austenitic 1.5mm thickness (16 gage) complying with BS 1449: Part 2 : 1983 and BS 4360. Mitred and welded units.

- C. Fully concealed Promaseal intumescent seal door strips, or equal approved.
- D. Acoustic seals shall be manufactured by Raven or equivalent to achieve the specified acoustic ratings and approved by the Architect. (Ref: Lorient Le 1212 and RP8). Refer to door schedule for locations of doors with acoustic seals.
- E. Welded joints shall be fully fused throughout with no holes, pores or cracks. Butt and fillet welds visible in the completed work shall be ground flush and smooth.
- F. Water bars shall be welded to the frames for the complete width of the doors.
- G. Flashing and weather bar shall be welded to frames for the complete width of the doors, when specified.
- H. Weather Seal: Provide at all exterior doors, jamb and head: Steelcraft Weatherstrip or equal approved.
- I. Weather Seal: Provide at all exterior doors sills: PS-074, cill-steelcraft bottom seal, FAS or equal approved.
- J. Finishes
 - 1. Finishes to frames of steel louvres and doors shall be either :-
 - 2. Sprayed zinc coatings to B.S. EN 22063 : 1994 with a minimum coating of 0.2mm thick,
 - 3. Hot-dip galvanized coatings to B.S. 729 : 1971 (1994) and with a minimum coating mass of 120 g/m² for doorset (460 g/m² for handrail use), or
 - 4. Powder coated to B.S. 6497 : 1984 (1991), after hot-dip galvanizing.
 - 5. Steel louvres, including frames, clips and weatherstrips shall be hot dip galvanized to B.S. 729 : 1971 (1994).
 - 6. Protective coatings are to be smooth, continuous and free from flux stains or other defects.
 - 7. No hot dip galvanized or zinc sprayed components are to be welded or drilled and any subsequent cutting is defined as damage which must be made good.
 - 8. Make good damage to hot dipped galvanized or zinc-sprayed coatings, treating cut or otherwise damaged galvanized components with two coats (minimum) of metallic zinc-rich priming paint to B.S. 4652 : 1995 : Type 2.
 - 9. Final finishes on steel components (undercoats and finishing coats etc.) shall be as described in Section 8.3 PAINTING or as shown on drawings.
 - 10. Where practicable, components should be supplied to site fully finished. Where this is not practicable, all galvanized components shall be applied with two (2) coats of approved zinc chromate based etching primer before delivery to site.
 - 11. Apply strippable protective tape over all surfaces prior to delivery of components to site.
 - 12. Provide temporary steel clamps fixed at the top and bottom of all opening lights prior to transportation to site.
 - 13. Wrap all components and fixings in stout polythene or other waterproof covering to protect against damp and scratches.

2.02 ANCILLARY MATERIAL

A. COUPLING MULLIONS AND TRANSOMES

1. Coupling mullions and transoms shall be provided as indicated on drawings and:-
 - a. of total length to suit door openings
 - b. Extending 75mm at each end for building-in,
 - c. Completed with all necessary coupling screws and nuts at not more than 900mm centres, commencing not more than 190mm or less than 50mm from the end of the frames,
 - d. Bedded in mastic with all interstices completely filled.

B. GLAZING BEADS

1. Refer to PS-A015 section 3.17 for fire rated glazing. Unless specified otherwise, steel doors shall be manufactured for internal glazing with beads.
2. Glazing beads shall be :-
 - a. Rolled steel heavy channel section,
 - b. Size 20 x 10mm for doors,
 - c. Mitred at all corners,
 - d. Fixed with galvanised mild steel flat headed counter-sunk screws at 225mm (maximum) centres, including tapping frames to receive screws, and
 - e. Structurally adequate and fixed to a sufficient number of points to withstand the design wind loading.

C. WEATHERSTRIP

1. Weatherstrip, when specified for opening doors shall be neoprene or polyvinyl chloride gaskets complying with B.S. 4255 : Part 1 : 1986 (1992) or nylon pile weatherstrip.
2. Weatherstrip shall be securely fixed into the dovetailed grooves in the sections to provide a continuous contact between the opening and the fixed frame.

D. SEALANTS

1. All sealants to be used in the louvre system shall be high range sealants, which are capable of performing well under dynamic loads. Maximum cyclic movement capability shall be from $\pm 12.5\%$ to $\pm 25\%$.
2. All sealants shall be identified and located by product name on shop drawings. Joint cleaner and primers shall be to manufacturer's recommendations.
3. The Architect shall select the colour from the manufacturer's range of standard colours.
4. Unless otherwise recommended by the sealant manufacturer, backer rods or back up material shall be polyethylene foam sponge neoprene conforming to ASTM C509 or open cell polyurethane foam rods, density 4-6 pcf, non-gassing, non-staining, non-wicking and non-hygroscopic. Diameter size of the rod to be at least 25% larger than the joint width.
5. The products offered must be compatible with the characteristics and properties of the contacting and neighbouring surfaces.

E. GROUTING

1. Waterproof cement grouting around frames shall consist of a mix in the following proportion :-
 - a. 50 kg cement,
 - b. 150 kg clean washed sand,
 - c. 5 kg waterproof compound.
2. Waterproofing additive for grouting shall be approved proprietary type to be applied in strict accordance with the manufacturer's instructions.

F. FIXING LUGS AND SCREWS

1. All steel louvres and doors shall be provided with steel fixing lugs and counter-sunk screws spacing at not more than 900mm centres, commencing not more than 190mm or less than 50mm from the end of all outer frames.
2. Fixing lugs shall be :-
 - a. Slotted adjustable lugs and screws necessary for building in the steel doors.
 - b. Protected at least 60mm beyond the metal frames and built into the structural openings.
3. Holes for fixing lugs or screws shall be driven at the centre of the web.

G. SLIDING DOORS

1. Sliding doors shall be constructed in such a way that the sliding panel cannot be removed from exterior when in a locked position.
2. The sill shall be so designed that water will run off to the outside.
3. Self-tightening interlock shall be provided on meeting stiles.
4. Sliding doors shall be provided with :-
 - a. Overhead rolling hanger and track with brackets,
 - b. Bottom roller and guide channel,
 - c. Fixed head, jamb and sill including weatherstrip,
 - d. Flush pulls, flush bolts and locking device.

H. ADJUSTABLE LOUVRE FRAMES

1. Frames and louvre clip of adjustable louvres shall be of not less than 1.2mm and 0.9mm galvanized steel sheet respectively.
2. Blades holders to be zinc alloy die cast, pivoted along the blade's centre line.
3. Clips shall be suitable for receiving 100mm or 150mm wide and 6mm thick glass blades, as required.
4. Weatherstrips shall be of shaped metal strips, of an approved design & thickness.
5. Steel frames, clips and weatherstrips shall be hot-dip galvanized or electro-zinc plated on all surfaces prior to assembly.
6. Frames shall be designed for hand or pole operation, as specified, with single lever control and to automatically lock when closed.
7. Provide where specified 45 x 10mm galvanized steel sub-frames for single side louvre frames, complete with slotted adjustable lugs and screws for building in.
8. Where required, matching rustless screws, bolts and spacers, etc. shall be provided for coupling adjacent frames to form mullions.

9. All fixings supplied are to be non-ferrous or suitably coated as to be free from rust in use.

I. STEEL FIXED LOUVRES

1. Frames shall be galvanized steel, "Z" shaped or angle sections, of the size indicated on drawings.
2. Louvre blades shall be :-
 - a. 1.5mm thick, unless otherwise specified,
 - b. 3mm thick for transformer room,
 - c. Bent to shapes as shown on drawing.
3. Steel fixed louvres shall be constructed to be weathertight and fixed at locations as shown on drawings.

J. METAL FRAMES

All door frames shall be Grade II Heavy-Duty Level capable of meeting the door tests of ANSI A250.4-1994 for operating cycles and twist tests.

1. Materials
 - a. Steel: Paint quality cold-rolled, pickled and annealed, carbon steel sheets conforming to ASTM A366 with ASTM A653 hot dipped A60 galvanized coating. Hot-rolled steel sheets are not acceptable.
 - b. Stainless Steel: Type 316 for applications, in hairline finish in accordance with ASTM A240/A240M-04a.
2. Fabrication
 - a. Minimum steel or stainless steel thicknesses:
 - 1) Interior frames: 1.5mm
 - 2) Exterior frames: 2.0mm
 - 3) Hardware reinforcing: Butt hinges and pivots: 3.0mm; Locks: 1.6mm; Surface-applied hardware: 2.0mm
 - 4) Dust covers (plaster or mortar guards) for concealed reinforcement: 1.0mm.
 - b. Die-cut or saw miter, or cope and tab all frame miters to provide a secured interlocking joint.
 - c. Frame shall have mitered corners and shall be of knockdown construction. Knockdown frames shall have self-aligning tabs and slots for securely locked corners.
 - d. Prepare jamb rabbet stops for "punched" rubber door silencers. "Stick-on" type silencers are not acceptable. Locate three for single doors and two for pairs of doors.
 - e. Provide concealed reinforcement to receive mortise type hardware. Mortise, drill, and tap to template requirements.
 - f. Provide spot or projection welds and 1) arc welds, or 2) 3.0mm thick "Z" shaped hinge reinforcement strap with at least three (two to frame and one to reinforcement) arc welds, above and below cut-outs for all butt, pivot, and floor hinges.
 - g. Reinforce frames for closers.
 - h. Provide stainless steel components and hardware reinforcements for stainless steel frames. Galvanized or mild steel reinforcing will not be accepted.

3. Frame Anchors: Same gauge as frame material.
 - a. For frames in stud partitions:
 - 1) Factory-attached (i.e., welded), non-adjustable, galvanized steel, of appropriate type for specific installation conditions
 - 2) Provide floor anchors in addition to specified jamb anchors
 - b. For frames integrally built with masonry walls: Loose "tee-shaped" or factory-attached (i.e., welded) adjustable galvanized steel anchor with perforated or crimped strap.
 - c. For frames installed subsequent to construction of concrete and masonry walls (including existing concrete and masonry): Combination bolt and spreader anchor consisting of a 50mm wide flat galvanized steel spreader plate welded to a pipe or tube spacer. "Butterfly" or similar plate-type anchors that obstruct flow of grout are not acceptable. Factory-attach (i.e., weld) anchor to frame; weld spacer to center of door stop and weld spreader plate to back-bend (return) flanges of frame. Capping shall be in same material and finish as frames.
 - d. Provide not less than three anchors for each jamb. Provide additional jamb anchors for door frames in excess of 2300mm height, in accord with HMMA 861, ANSI A250.8 (SDI 100) or as recommended by manufacturer. All door frames and frame anchors shall be tested and meet the requirement of ANSI A250.5-1994.
4. Finish painting to steel: Clean galvanized surfaces and bake one coat of polyester powder coating to a minimum dry film thickness of 60µm in accordance with ANSI A250.10. (Ral color 7040).
5. Stainless steel frames to be self-finish in hairline grit, without any scratches or distortions on facings.
6. Grout mix for frames installed subsequent to construction of concrete and masonry walls.
7. Leaf height over 2230mm need to use 4 nos. of s/s ball bearing hinges.
8. For more details, please refer to door schedule.

K. PROVISION FOR EQUIPOTENTIAL BONDING

1. Provide all external metalwork with screw holes or other connection points to the approval of the Architect for the purposes of connection, by the Electrical Installation Sub-Contractor, of equipotential bonding conductors as required by the I.E.E. Wiring Regulations.

2.03 FABRICATION

- A. Fabricate all steel doors sets to the following tolerances.
 - 1) Width tolerance 1mm
 - 2) Height tolerance 2mm
 - 3) Smoothness max. 0.15mm
 - 4) Squareness max. 0.1
- B. All steel doors shall be rigidly built and free from warp and buckle. The sheet shall be flat and completely free from any telegraphing of welds.
- C. Select core materials from clause 2.02 above as required to meet the required fire or acoustic performance, or manufacturer's equivalent, subject to Architects approval.

- D. Fabricate frames with all required hardware reinforcement plates welded in place. Provide mortar guard boxes. Prepare frames for all silencers.
- E. Reinforce units to receive both surface applied and mortise finish hardware. All ironmongery is to be fully mortised into frame unless noted otherwise.
- F. Neatly prepare and all steel doors to receive ironmongery, intumescent and smoke seals, and weatherstripping provided under Section 08710.
- G. Prime and finish all cut-outs, recesses and grooves for door ironmongery, intumescent and smoke seals, and weatherstripping. The colour of the finish is to match the laminate colour.
- H. Under cut non-fire rated doors as indicated on the Drawings. Install door bottom 'V' channel at all under cut doors per. Ensure weatherproof installation.
- I. All door ironmongery and hardware shall be fully rebated into door. Surface mounted ironmongery or other hardware will not be permitted.
- J. Acoustic Door Sets:
 - 1. Provide in accordance with the Acoustics Consultants Specifications and door schedule.
 - 2. Seals and sealing mechanisms shall be adequately protected against damage during transit and installation. Site cutting of seals shall be limited to an absolute minimum, and shall be approved by the Architect.
 - 3. Threshold sealing mechanisms shall be adjustable, to allow for the accommodation of door movement and normal building tolerance without loss of performance, but the seals shall not be used to take up tolerances greater than +/- 30% of its design dimension without being overstretched. The threshold seal shall consist of a compression seal designed for use with an associated threshold limit. Compression seals shall be fitted to the frames of all doors to form continuous seals around the jambs and heads.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify substrate conditions, opening sizes and tolerances are acceptable for proper installation.
- B. Verify that door frame has been properly installed to receive door.

3.02 INSTALLATION

- A. Allow adequate time for all doors to condition and acclimatize to their psychometric environment prior to installation.
- B. Install doors in accordance with fire rating approval requirements and Manufacturer's instructions.
- C. Coordinate installation of doors with installation of ironmongery, intumescent and smoke seals, and weather strippings as specified in this Contract.
- D. Doors to finished ready to receive finish painting under Section 8.3 PAINTING work.
- E. Set frame within opening, plumb jambs and level head. Brace frame as required. Build frame into wall. Fully fill the frame with mortar as the work progresses. The frame alignment is to be checked as the wall is built.
- F. Coordinate with adjacent wall construction for anchor placement.
- G. Provide mortar filling of door frames at all fire-rated doors and at locations indicated on the Drawings. The voids between door frames and wall shall be carefully packed back and made good with proper caulking, sealant and plaster.
- H. All frames and architraves shall be fully sealed to the structure with mortar or plaster. Gaps not exceeding 6mm may alternatively be sealed with a non-setting polysulphide or silicone sealant.
- I. No light shall be visible through any installed doorsets when observed in darkness from either side, with room lighting on the opposite side. Where specified in the Drawings or Schedules, doors shall be fitted with an automatic door closer to eliminate slamming noise.
- J. Where specified in the Drawings or Schedules, compression seals shall be fitted to the door rebate sides and head, and shall meet the door leaf at all points when the door is closed.
- K. Seals shall not be stretched.
- L. The door seals shall not inhibit normal use of the door or make the door difficult to close.
- M. Where specified in the Drawings or Schedules, perimeter seals and bottom drop seals shall be provided as a standard integrated fixture of the door assembly. A drop type threshold seal shall be rebated in to the bottom of the door. This shall be activated by door closure.
- N. Intumescent fire seals shall be provided in addition if required, as specified by the Architect.

- O. Each individually installed door set shall provide the specified minimum acoustic sound ratings, as measured in accordance with the latest revisions of ASTM E90-85 and ASTM E413-87, or their equivalent.
- P. Acoustic door sets shall be installed in general accordance with the following installation sequence:
1. Check and confirm the sizes of structural opening and door opening against the door schedule.
 2. The voids between door frames and wall shall be carefully packed back and made good with proper caulking, sealant and plaster.
 3. Fix the door frame in place and leave all fixings loose.
 4. The door leaf shall now be hung and adjusted such that the seals on the hinge post side are correctly functioning. With the door closed, the frame should be adjusted up to the door blank, ensuring that there is no undue strain between the blank and the frame. The gap at the top of the door shall be uniform and parallel, achieved if necessary by packing up the threshold.
 5. When satisfied with the adjustment, the frame should now be fixed in place, both horizontally and vertically, using packing pieces where necessary, so as not to destroy the regular opening between the frame and the leaf.
 6. The striker plate can now be adjusted away from the door blank such that there is sufficient clearance to avoid rubbing when the door is opened or closed.
 7. Where appropriate, the threshold can now be packed up and fixed, ensuring that there is no strain or bow introduced and the latch plate must now receive its final adjustment such that the door compresses the seals evenly around the perimeter.
 8. All frames and architraves shall then be fully sealed to the structure with mortar or plaster.
 9. Notwithstanding the guidance on installation given above, the Contractor must ensure full compliance with the required acoustic ratings given of this Section and the Specification for Acoustic Performance of Acoustical Doors.
- Q. Fixing Steel Doors
1. Handle and store components on site in strict accordance with manufacturer's instructions. Avoid damage and distortion of frames and protect all decorative finishes against damp, scratches, knocks etc. Maintain protective polythene or other wrappings on components for as long as practicable.
 2. Assemble composite components, using all necessary mullion and transom couplings, bolts, spacers, screws etc. Bed mullions and transoms in mastic and seal all joints with one-pack gun grade polysulphic sealant to B.S. 5215 : 1986. Ensure that all interstices are completely filled to provide fully weathertight joints.
 3. Paint one coat of bituminous paint on concealed steel surfaces which will be or may come in to contact with cement, wet mortar, plaster or similar materials and finishes.
 4. Position components plumb, level and square in openings. Form pockets at head, jambs, cills etc. and fix and build in lugs and screw to frames; or plug and screw fix frames using packing pieces as necessary.
 5. Avoid distortion of frames. Care shall be taken to ensure that fixing points are aligned correctly between frame and background. Do not over tighten bolts or screws causing localised distortion of the frame at fixing points.
 6. Bed frames in waterproof cement sand grout 1:1 leaving no gaps. Rake out joints around the external edge of frames to the approval of the Architect. Mask adjoining surfaces from which it would be difficult to clean off mastic. Clean

joint and point with an approved oil based mastic sealant, finished to a smooth flat fully waterproof joint.

7. Remove excess pointing material and sealant from adjoining surfaces and leave clean.
8. Remove strippable coatings immediately prior to handover inspection. Remove surplus adhesive from surfaces with a suitable non-damaging solvent approved by the manufacturer. Wash down surfaces with clean water to remove any residues.
9. Lubricate and adjust all operating mechanisms leaving clean and in perfect working order on completion.

R. Fixing Steel Louvres

1. Handle and store louvres on site in strict accordance with manufacturer's instructions. Avoid any damage or distortion to frames or decorative finishes. Maintain protective wrapping for as long as practicable.
2. Fix louvres, level and square in openings. Bed frames in waterproof cement and sand grout (1:1) leaving no gaps.
3. Fix louvres as specified and either :
 - a. plug and screw fix sub-frames, packing out as required, or form pockets around sub-frames for fixing lugs. Build in lugs, making good to opening, and screw fixing lugs to sub-frames.
 - b. Where required, plug and screw fix louvre frames and weatherstrips direct to opening.
4. Where required, screw single sided louvre frames to sub-framed at 225mm centres. Avoid distortion of frames when tightening fixings.
5. Paint one coat of bituminous paint on concealed steel surface.
6. Rake out and point external edge around frames in oil based mastic.
7. Remove strippable coatings and clean down frames.

3.03 CO-ORDINATION

- A. Coordinate adjustment of doors with installation of ironmongery.

3.04 CLEANING

- A. Clean all steel doors free from dirt, dust and stains. Doors shall neither squeak nor rattle when operated.

3.05 PROTECTION

- A. After steel door assemblies are installed, The Contractor shall properly and adequately protect the steel doors from damage. Suitable cushioned protection shall be provided wherever required, which shall not damage or deface doors, door frames or fittings.
- B. All steel door set work in progress shall be protected at all times during construction by use of a suitable strong, impervious protective film and board securely held in place and maintained. Additional protection shall be provided by the Contractor at heavily trafficked areas, and locations subject to impact/abuse.
- C. Remove protective film on door frames at time of Project handover to the Employer. Remove all finger prints and polish surfaces.
- D. The Contractor shall repair, or if necessary replace, all damaged steel doors and accessories to the satisfaction of the Architect.

3.06 ADJUSTMENT

- A. Test for smooth operation through full range of door swing; make necessary adjustments.
- B. Adjust doors and ironmongery for smooth and balanced door movement, and shall neither squeak nor rattle when operated.

END OF SECTION